The equations of motion may be expressed in terms of the Lagrangian as:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\boldsymbol{q}}} = \frac{\partial L}{\partial \boldsymbol{q}}.$$

(Note that this must be interpreted in general as a set of $\mathrm{N_c} = 3\mathrm{N}$ equations, one for each atomic coordinate.) The real usefulness of the Lagrangian perspective comes when we introduce different types of coordinates to express the equations of motion; examples of this are given below.

When we introduce a smooth change of variables $\boldsymbol{q} = \boldsymbol{\Phi}(\boldsymbol{Q})$, where $\boldsymbol{\Phi} : \mathbb{R}^{\mathrm{N_c}} \to \mathbb{R}^{\mathrm{N_c}}$, this induces a corresponding transformation of the velocity vector by

$$\dot{\boldsymbol{q}} = \boldsymbol{\Phi}'(\boldsymbol{Q})\dot{\boldsymbol{Q}},$$

where $\boldsymbol{\Phi}'$ is the $\mathrm{N_c} \times \mathrm{N_c}$ Jacobian matrix of $\boldsymbol{\Phi}$. Then the Lagrangian of the system is transformed to

$$\tilde{L} = \frac{\dot{\boldsymbol{Q}}^T \boldsymbol{\Phi}'(\boldsymbol{Q})^T \boldsymbol{M} \boldsymbol{\Phi}'(\boldsymbol{Q})\dot{\boldsymbol{Q}}}{2} - U(\boldsymbol{\Phi}(\boldsymbol{Q})),$$

It is not essential for the change of variables to be a map from $\mathbb{R}^{\mathrm{N_c}}$ to $\mathbb{R}^{\mathrm{N_c}}$. For example, in a system with constraints, the change of variables might represent a parameterization of the manifold defined by the constraints. Then the mapping $\boldsymbol{\Phi}$ might be from $\mathbb{R}^{\mathrm{N_d}}$ to $\mathbb{R}^{\mathrm{N_c}}$, where $\mathrm{N_c} \geq \mathrm{N_d}$. An abstract set of configuration variables is referred to as a set of *generalized coordinates*. The introduction of generalized coordinates in an N-body system typically leads to a Lagrangian which can be viewed as quadratic in the velocities but with a "configuration-dependent mass matrix." We will assume that any such changes of variables are regular transformations in the sense that $\boldsymbol{\Phi}'$ is of full rank and the resulting generalized mass matrix is invertible.

Although generalized coordinates are often useful for understanding molecular systems, as we shall learn in later chapters, it is desirable to avoid the use of formulations with a configuration-dependent mass matrix since it complicates the design of numerical methods with good conservation properties. This is one reason that molecular simulation methods are typically described in a Cartesian coordinate framework.

1.4 Hamiltonian Formulation

In this section, we introduce the Hamiltonian description of the motion which is useful for discussing the underlying geometric structure associated to classical mechanics. We can understand the Hamiltonian formulation as a natural description of classical mechanics from several perspectives. First, it is based directly on the

most basic conservation law associated to a closed system (the conservation of energy) as it is formulated in terms of the energy function. Second, it is the starting point for the quantum mechanical treatment of a system of particles, since the Schrödinger equation is written in terms of the quantum analogue of the energy function. Third, we may view the Hamiltonian as the *Legendre transformation* of the Lagrangian viewed as a function of the velocities. This is a *conjugacy* relationship which ensures that the properties of the two systems are, under typical conditions, equivalent.

Abstractly, a Legendre transformation of a given convex function $g = g(\boldsymbol{\xi}) : \mathbb{R}^m \to \mathbb{R}$ is a new function $\tilde{g} = \tilde{g}(\boldsymbol{\eta}) : \mathbb{R}^m \to \mathbb{R}$ defined by

$$\tilde{g}(\boldsymbol{\eta}) = \sup_{\boldsymbol{\xi}} (\boldsymbol{\eta}^T \boldsymbol{\xi} - g(\boldsymbol{\xi})),$$

where the symbol $\sup_{\boldsymbol{\xi}}$ means to find the supremum of the given expression with respect to variation of the vector $\boldsymbol{\xi}$. In the case of the Lagrangian $L(\boldsymbol{q}, \boldsymbol{v}) = \boldsymbol{v}^T \boldsymbol{M}(\boldsymbol{q})\boldsymbol{v}/2 - U(\boldsymbol{q})$ (which we have seen is the formulation of a mechanical system in generalized coordinates) we find

$$\sup_{\boldsymbol{v}} (\boldsymbol{p}^T \boldsymbol{v} - (\boldsymbol{v}^T \boldsymbol{M}(\boldsymbol{q})\boldsymbol{v}/2 - U(\boldsymbol{q}))) = \sup_{\boldsymbol{v}} (\boldsymbol{p}^T \boldsymbol{v} - \boldsymbol{v}^T \boldsymbol{M}(\boldsymbol{q})\boldsymbol{v}/2) + U(\boldsymbol{q}).$$

If $\boldsymbol{M}(\boldsymbol{q})$ is invertible, the supremum is achieved precisely when

$$\boldsymbol{v} = \boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{p},$$

which gives the standard definition of the momentum vector $\boldsymbol{p}$ in terms of velocities,

$$\boldsymbol{p} = \frac{\partial L}{\partial \dot{\boldsymbol{q}}} = \boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{q}},$$

and, with $\boldsymbol{v} = \boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{p}$, the Legendre transformation results in a new function

$$H(\boldsymbol{q}, \boldsymbol{p}) \stackrel{\text{def}}{=} \boldsymbol{p}^T \boldsymbol{M}(\boldsymbol{q})^{-1} \boldsymbol{p}/2 + U(\boldsymbol{q}).$$

This is precisely the energy function, written in terms of positions and momenta. In this form it is termed the *Hamiltonian* and it provides an alternative description of the system. The equations of motion can be written

$$\dot{\boldsymbol{q}} = \frac{\partial H}{\partial \boldsymbol{p}}, \qquad \dot{\boldsymbol{p}} = -\frac{\partial H}{\partial \boldsymbol{q}}.$$

For constant $\boldsymbol{M}$ we obtain the dynamical equations,

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \qquad \dot{\boldsymbol{p}} = \boldsymbol{F} = -\partial U/\partial \boldsymbol{q}.$$